

ADAPTABLE AURAS

BY JAKE WELCH

Adaptable Auras is an installation designed to explore & challenge the role technology plays in the tradition of art by implementing reproduction & machine learning technologies to create an art piece that is intrinsically connected to those who view it.

Art contains the expression of the artist who created it. A unique piece of art contains something special that can only be truly felt in the original, an *aura*.

These elements are what connect us to other artists' work.

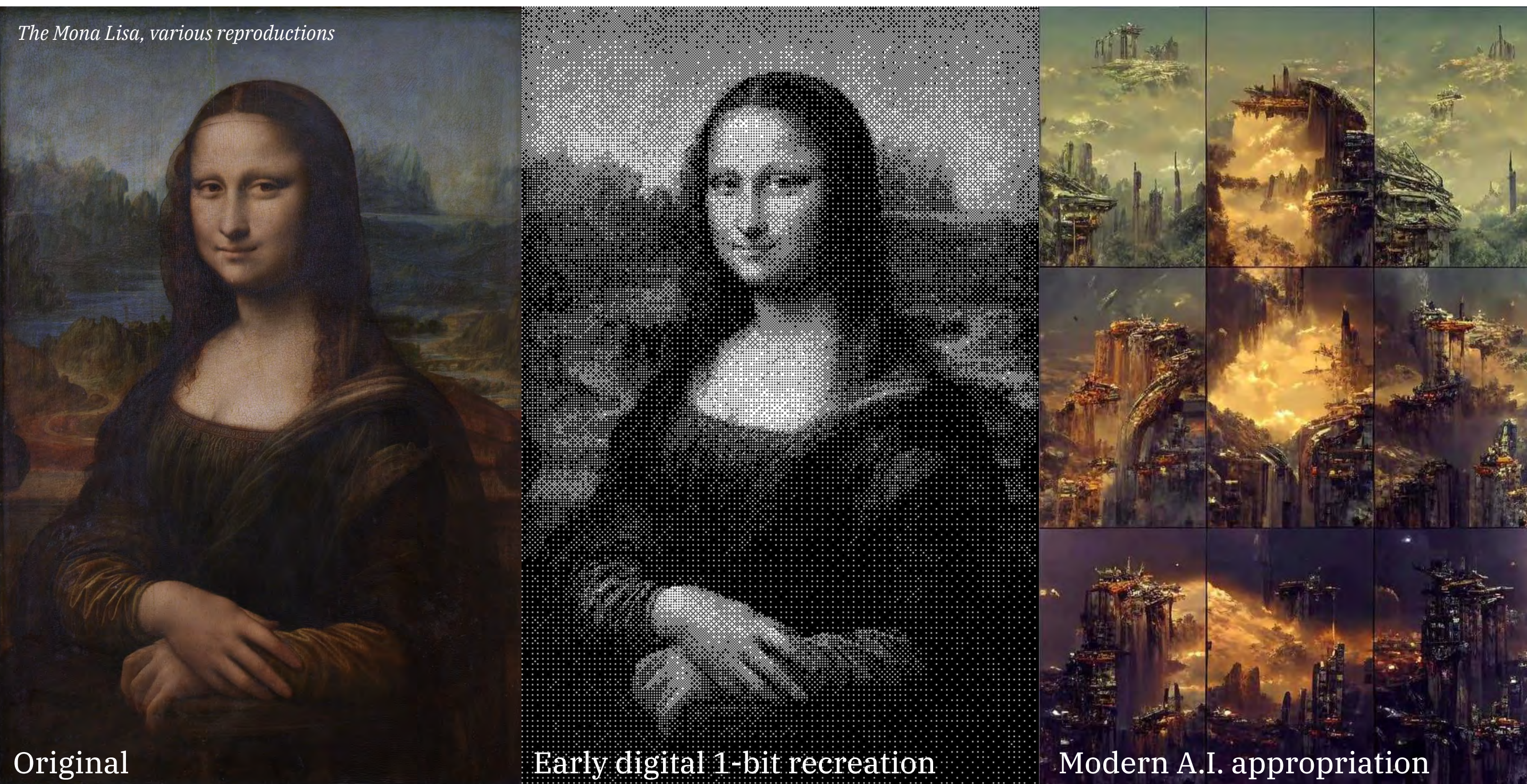
“What, then, is the aura? A strange tissue of space and time: the unique apparition of a distance, however near it may be. To follow with the eye – while resting on a summer afternoon – a mountain range on the horizon or a branch that casts its shadow on the beholder is to breathe the aura of those mountains, of that branch.”

- Walter Benjamin

Through the modern reproduction of art something of the original is lost. When an artwork is reproduced, its constant re-contextualization alters its meaning & diminishes its *aura*.



The Gallery of Cornelis van der Geest by Willem van Haecht, reproduced



With computers & the internet, it's now possible to digitally reproduce artworks easier than ever before.

With machine learning & AI image generation, famous artworks can be appropriated & reimagined by an algorithm, completely *removing the artist from the art*.

Technology is threatening to separate us from what makes art meaningful.

How can we use these technologies as tools to connect us to art rather than remove us from it?



From the beginning of this project, I have been deeply inspired & influenced by artists who use technology-centered installations to tell their stories.

I felt that creating an art installation would be the perfect way to approach this question.

Some early inspirations included the work of the artist Krzysztof Wodiczko who uses technology & projection to help tell his stories.

My preliminary research was spent examining technology-focused interactive art installations & using them to determine what I would & wouldn't want in my own project.

Precedent

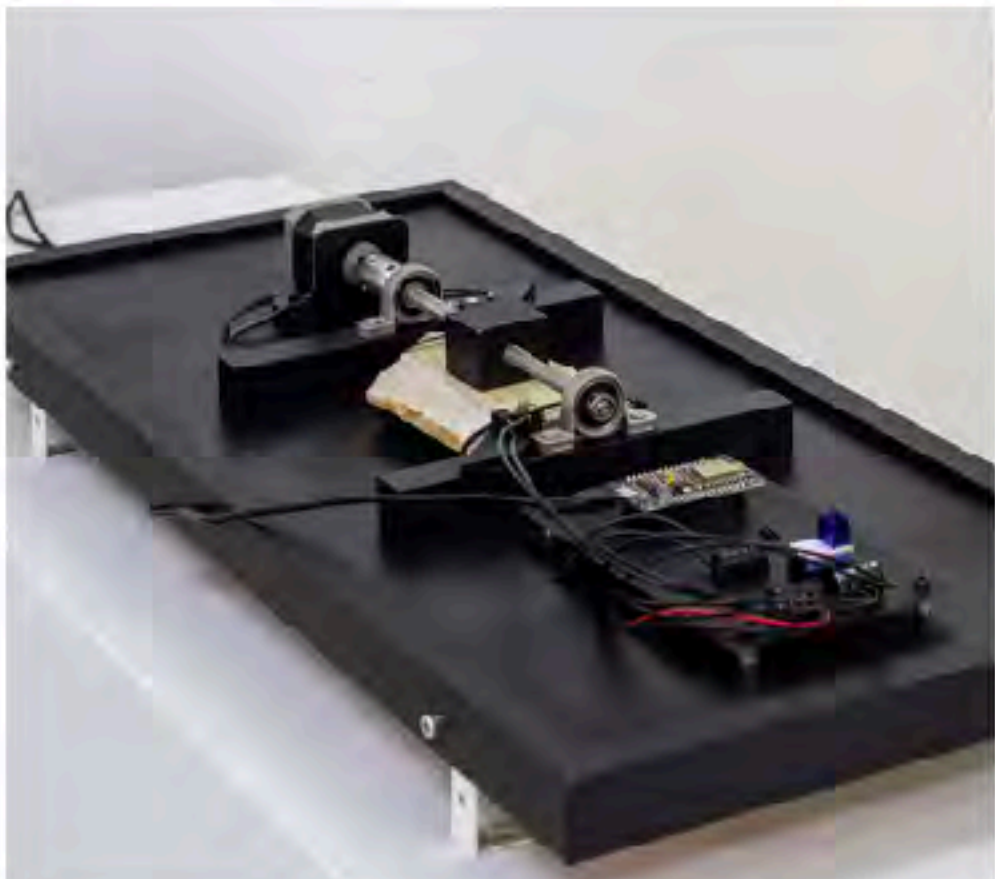
"Continuous Erosion" by Julian Hespeneide detects radio signals from guests' cell phones in the space to accelerate the movement of a machine which is moving to erode a physical object. This artifact tells a story of ecological conservation as well as the effects of the digital world on our minds & bodies.

Take

- > Mechanical installation which interacts with the audience without their knowledge
- > Tells a story of two separate ideas & elegantly ties the two together
- > I'm inspired by the material aesthetics of the exposed electronics

Leave

- > Raising awareness shouldn't be the main goal of the project



Precedent

"Ephemeral Data" by Jeroen van Loon was a performance art piece designed to visualize the ephemeral nature of data. The artist wrote: "The internet is evolving from an archive machine into a performance machine. Storing, saving and archiving copies of copies of copies is becoming obsolete. Temporary, local and unique content is becoming the new status quo." To achieve this, they had performers meticulously create a "mandala" of sand, the base material for creating fiber optic cables. At the end, it was all swept away never to be seen again.

Take

- > Physical art based on a digital phenomena
- > Material & nature of the piece informed by the narrative

Leave

- > Should remain autonomous, no human performers



Precedent

Michale Candy's "Persistence of Vision" utilizes tech & aesthetics commonly associated with surveillance to create an environment of play. It consists of computer vision powered spotlights made to look like CCTV surveillance cameras which follow people with a ray of light as they walk down an alleyway.

Take

- > Computer vision technology as the main data collection method
- > Interactive installation that encourages play

Leave

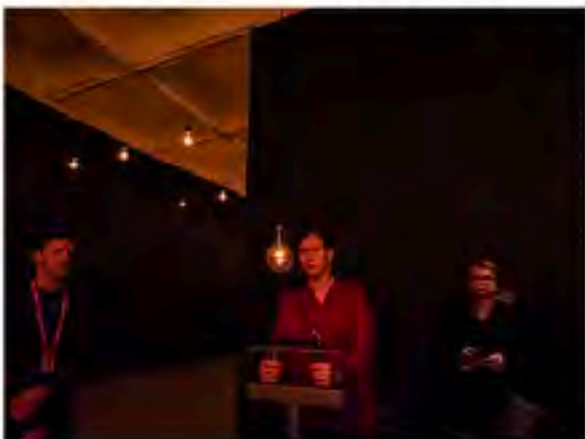
- > I don't want to focus on surveillance



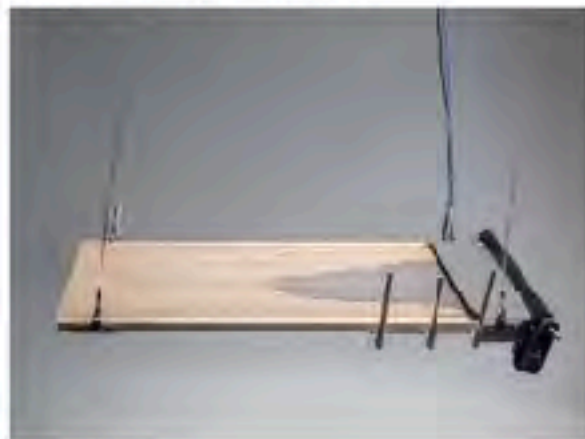
Other inspirations

- > Crowd interactivity
- > Bridge between physical & digital aesthetics
- > Digital manipulation of physical material
- > Human-centered data
- > The machine is the performer

Rafael Lozano-Hemmer, *Pulse Room*



Leonard Puhl, *Lowastream*



Petra Gemeinboeck, *ACCOMPLICE*



Golan Levin, *Dialtones (A Telesymphony)*



Soňa Taipa, *Evanescente*



Take:

- Mechanical installation
- Interactive
- Tells a story
- Uses computer vision
- Form is informed by the narrative
- Bridges digital & physical aesthetics
- Human-centered data
- Machine is a performer

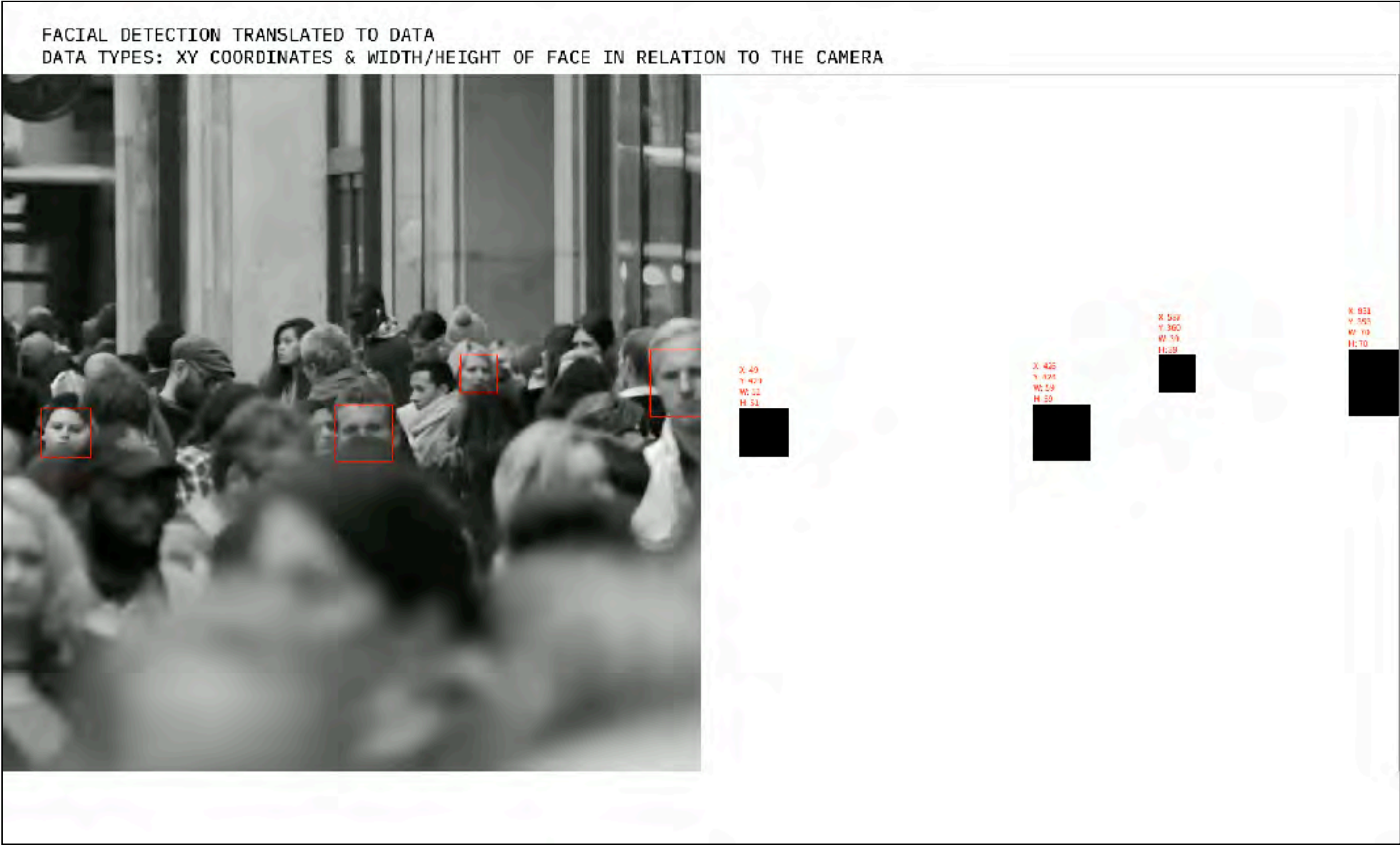
Leave:

- Raises awareness / activism
- Focuses on surveillance & privacy
- Depends on people to run it

Beginning this project required lots of experimentation & iterations as I haven't done most of this before.

Turning to tools I already know, my initial experiments focused on extracting data using computer vision tools in Processing.

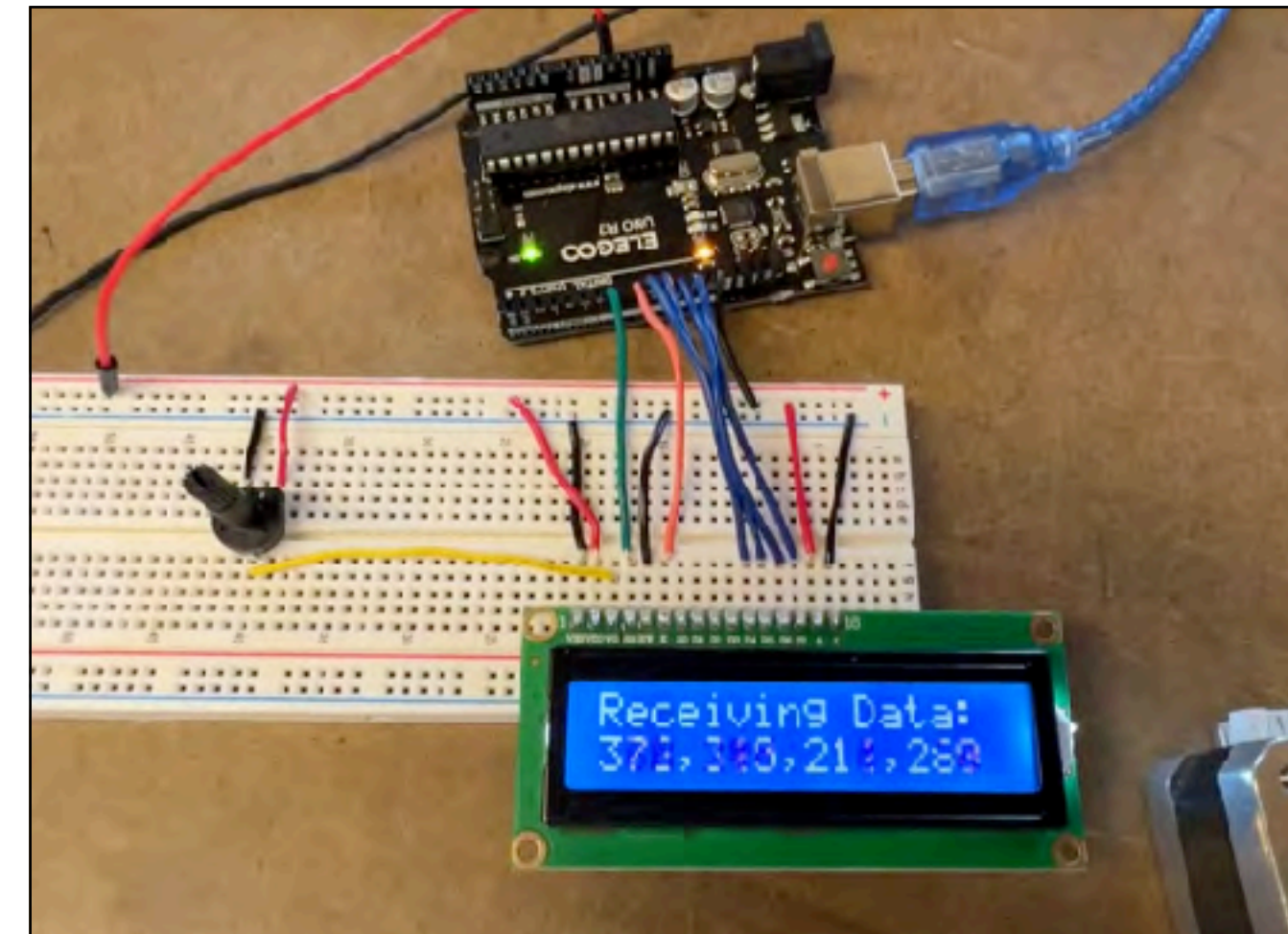
Here I am reading the X & Y coordinates & the width & height of every detected face in a frame.



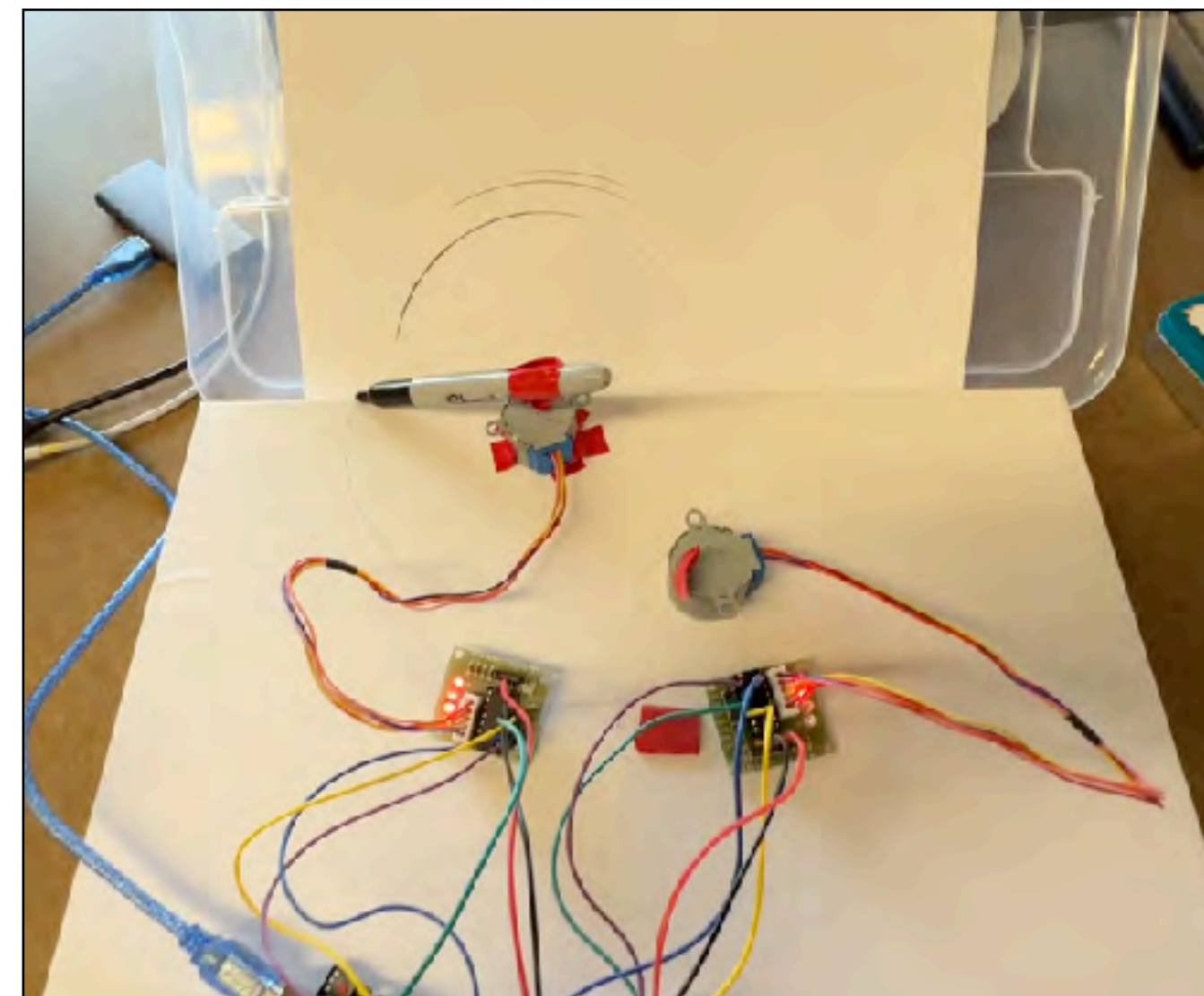
sketch_2023_08_23_Early_experiment.pde

As I began to consider how this data could be used in an installation, I ran tests to test if these values could be received by an Arduino.

It's success meant I could control electronics with an Arduino by sending it computer vision data, which I tested on some small stepper motors.



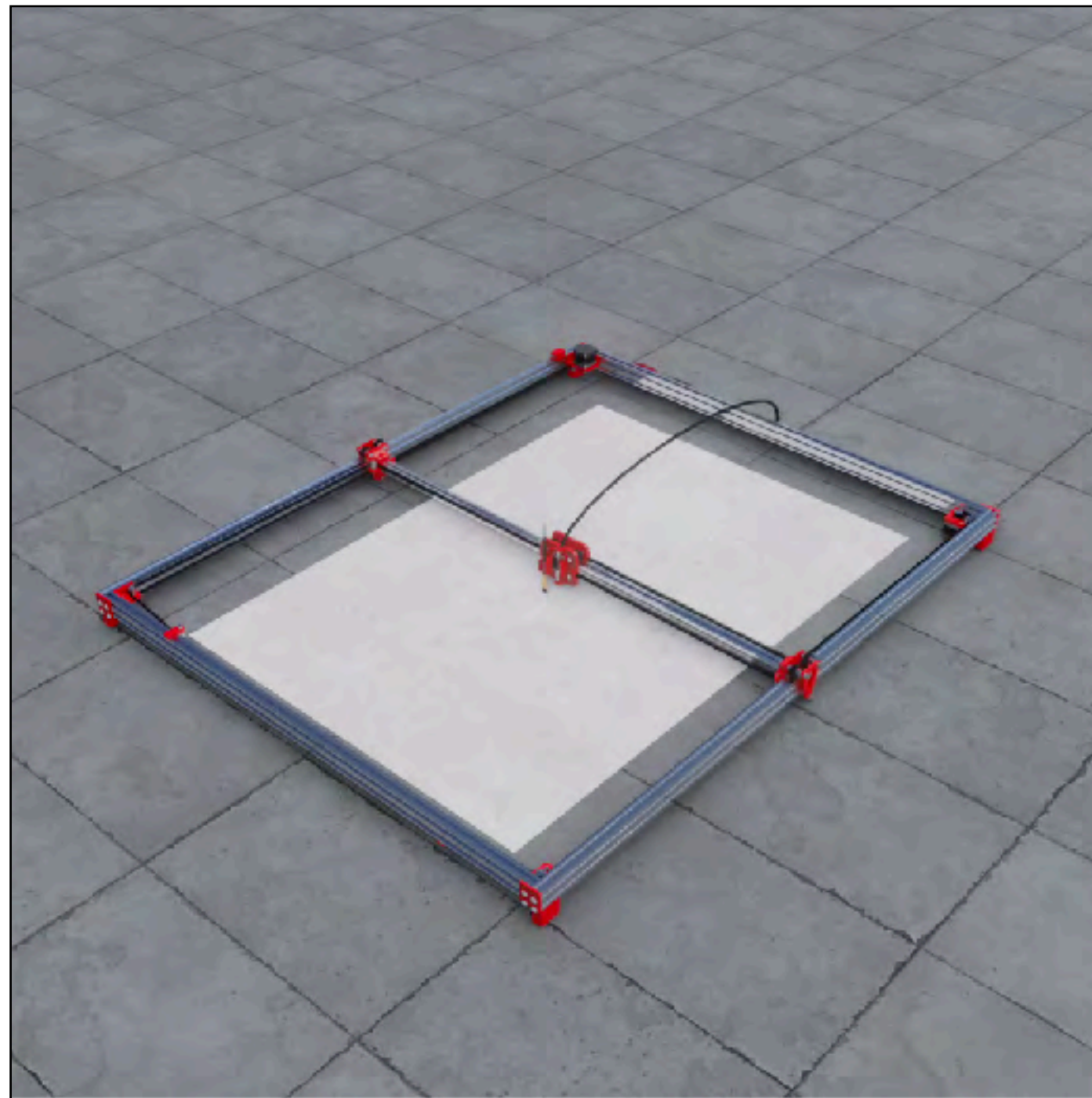
Sending CV data to Arduino via serial connection



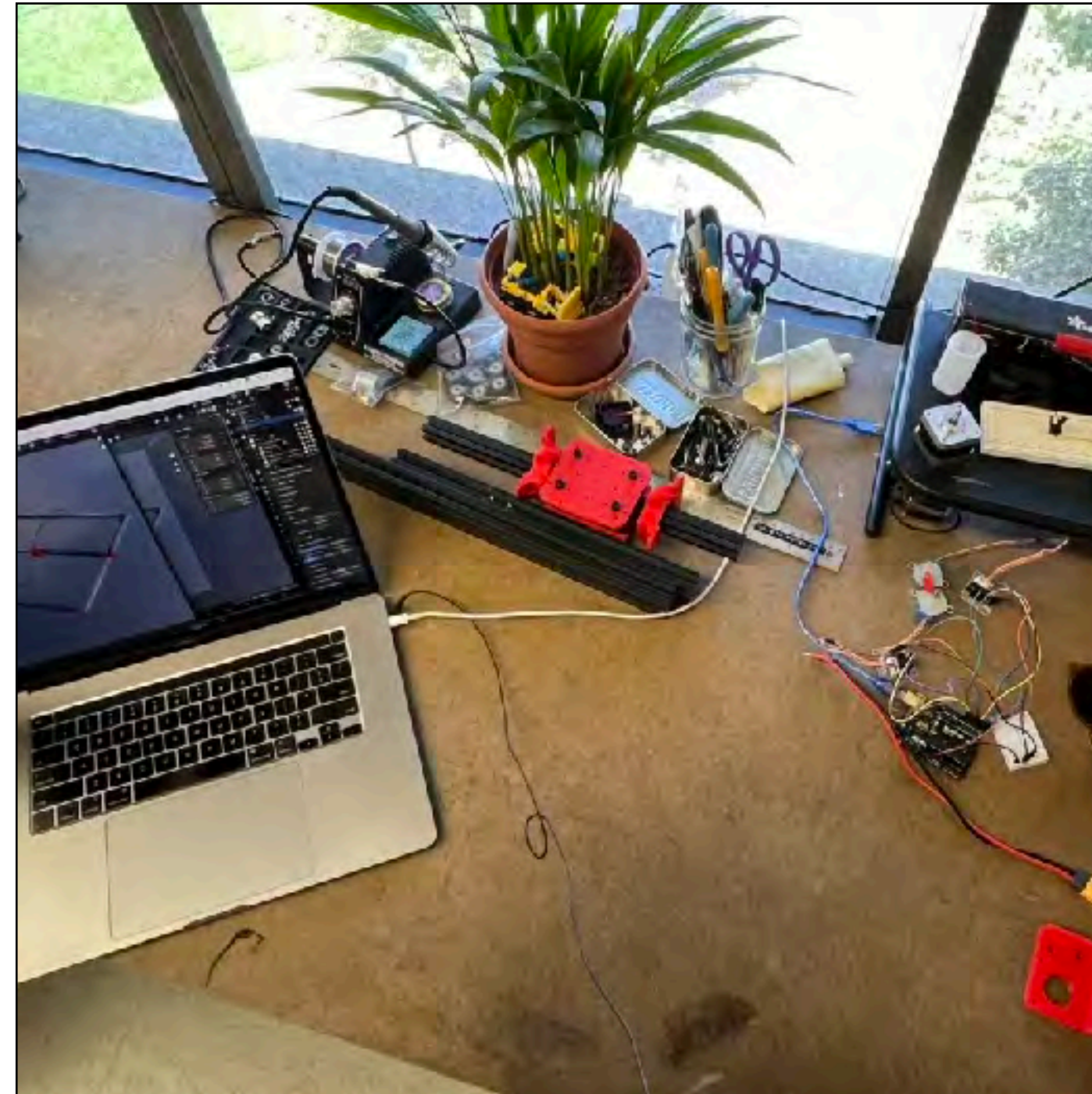
Using CV data to move stepper motors

With this proof of concept, I decided on building a pen plotter that could create draw with the data.

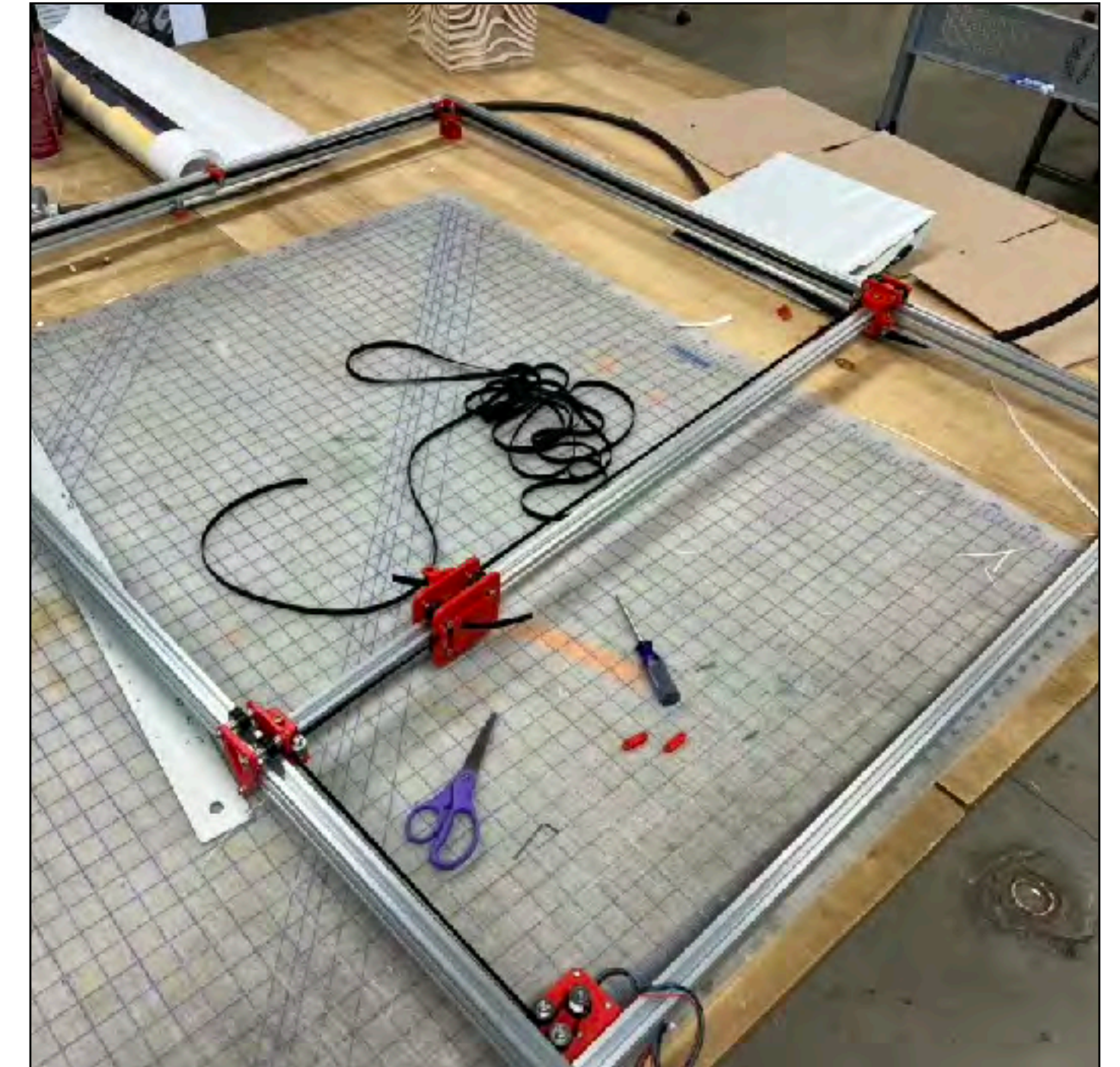
Following the “Caruther’s” coreXY plotter design, I 3D modeled the full build, 3D printed & purchased the parts, & began assembling all the components.



3D render of the plotter design



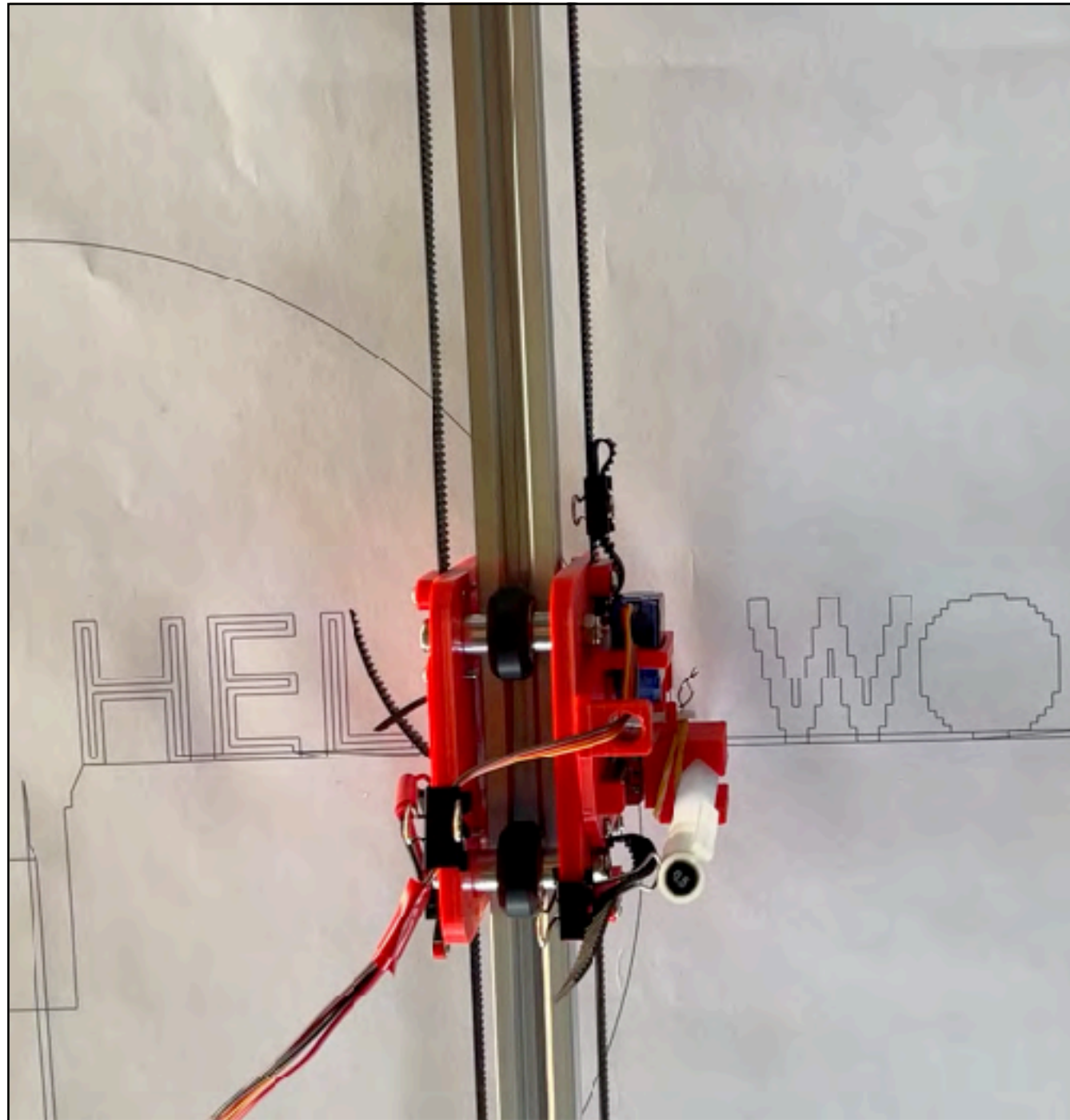
First 3D prints



Plotter assembly

Once built, several test drawings were made using existing CNC software to test the various components of the machine.

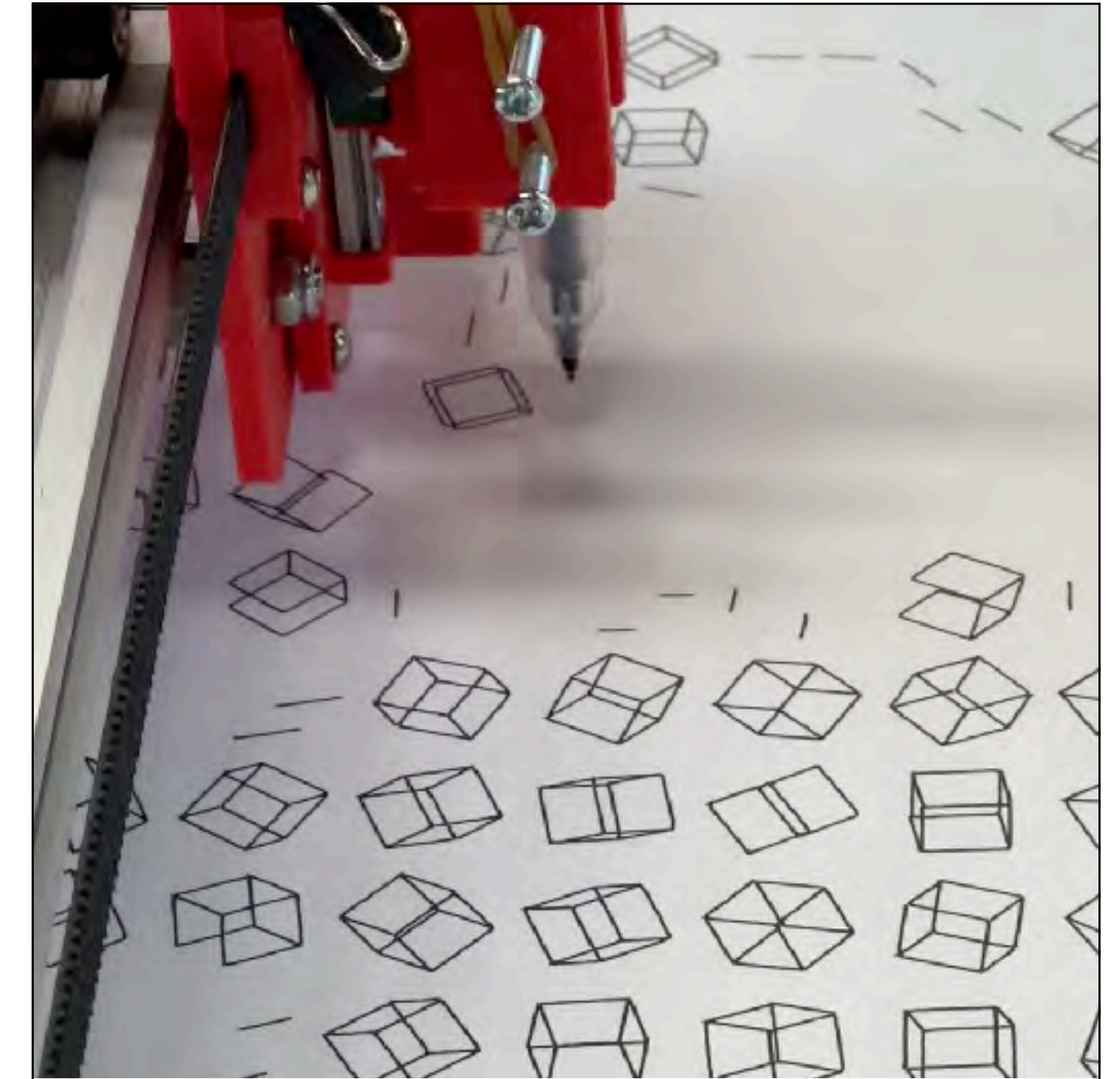
This laid the foundation for me to write my own program implementing the same commands.



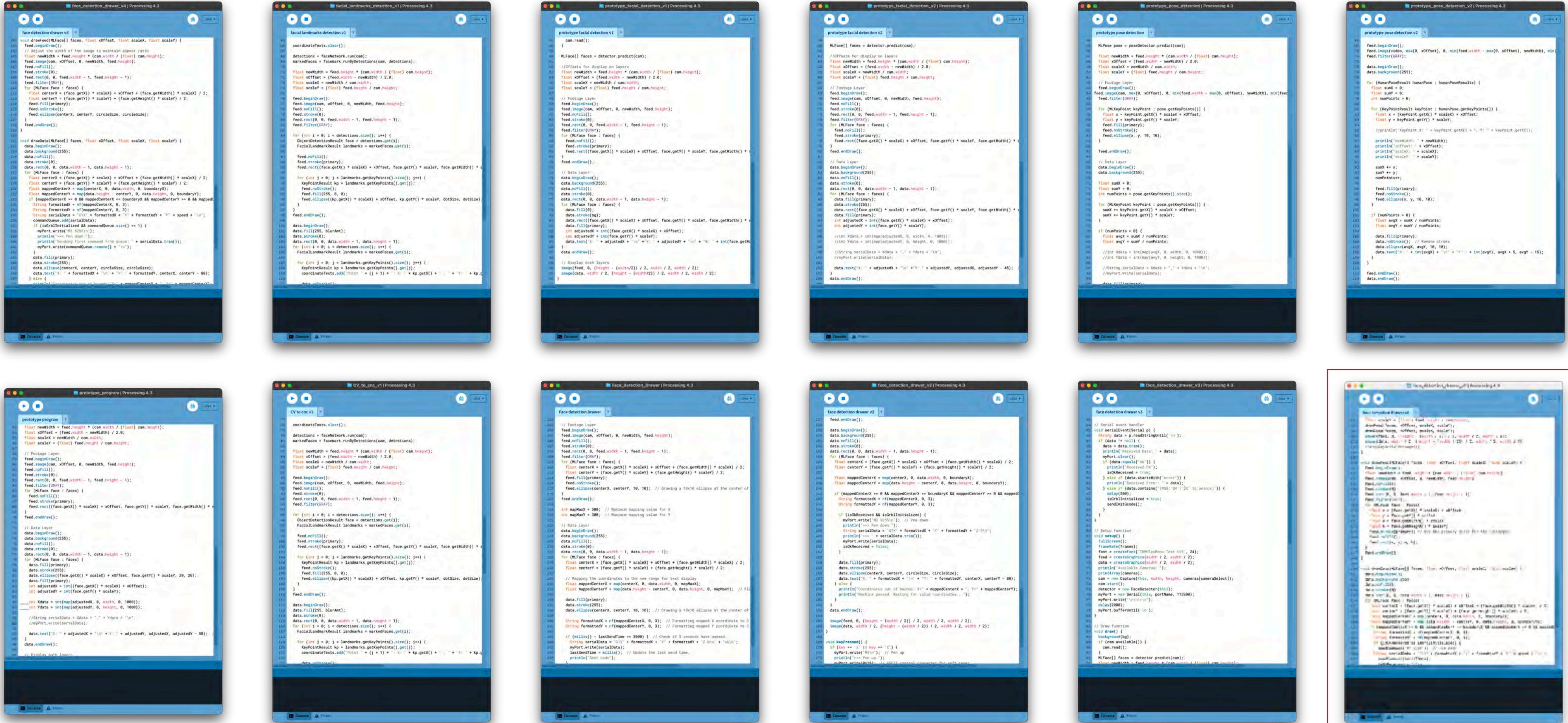
First successful drawing



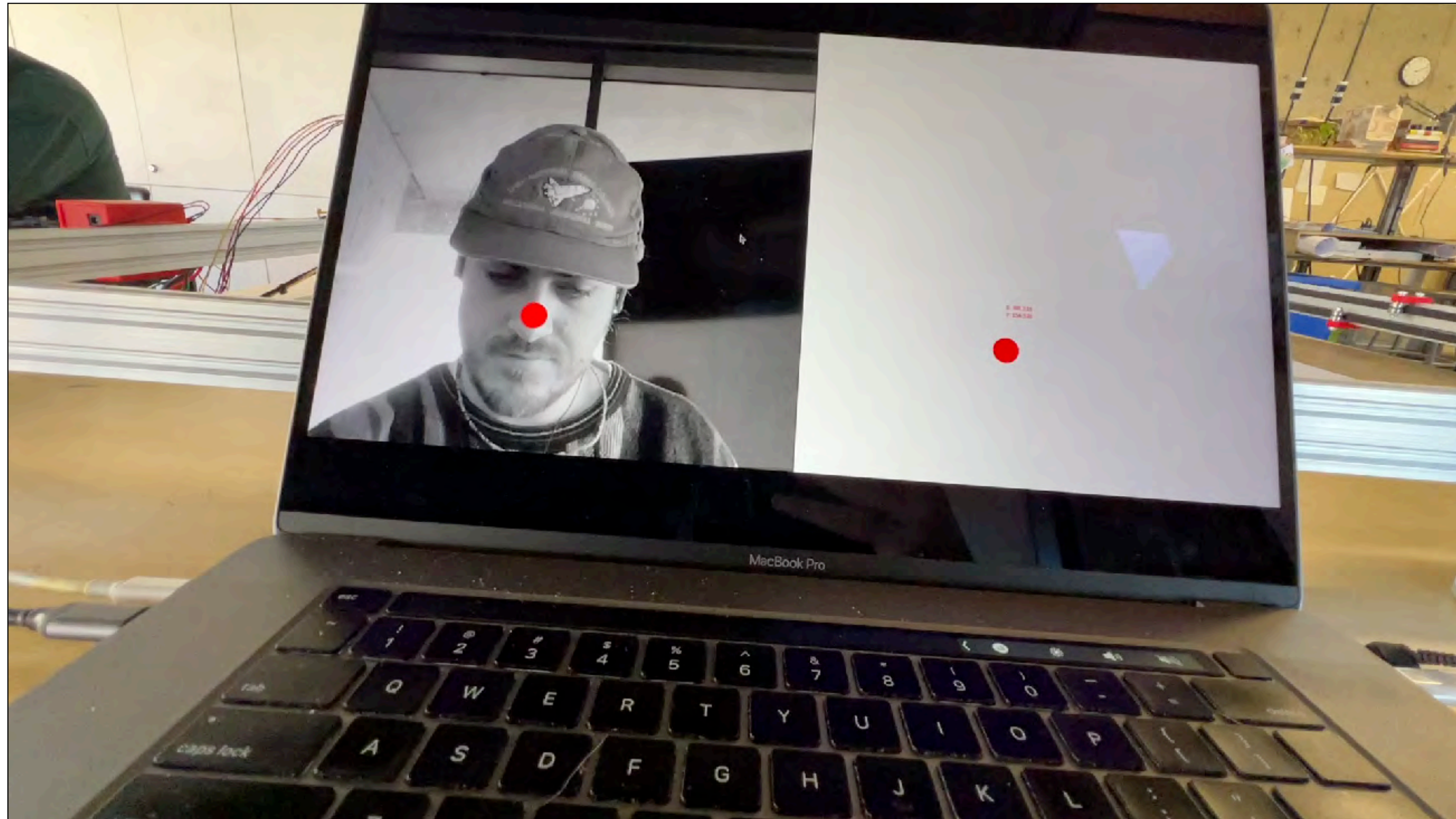
Pen speed test



Pen lift test

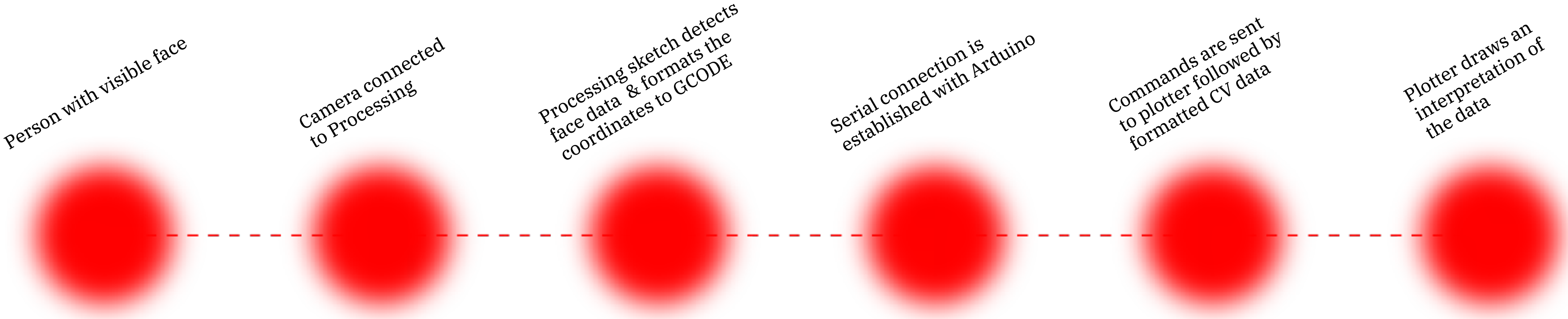


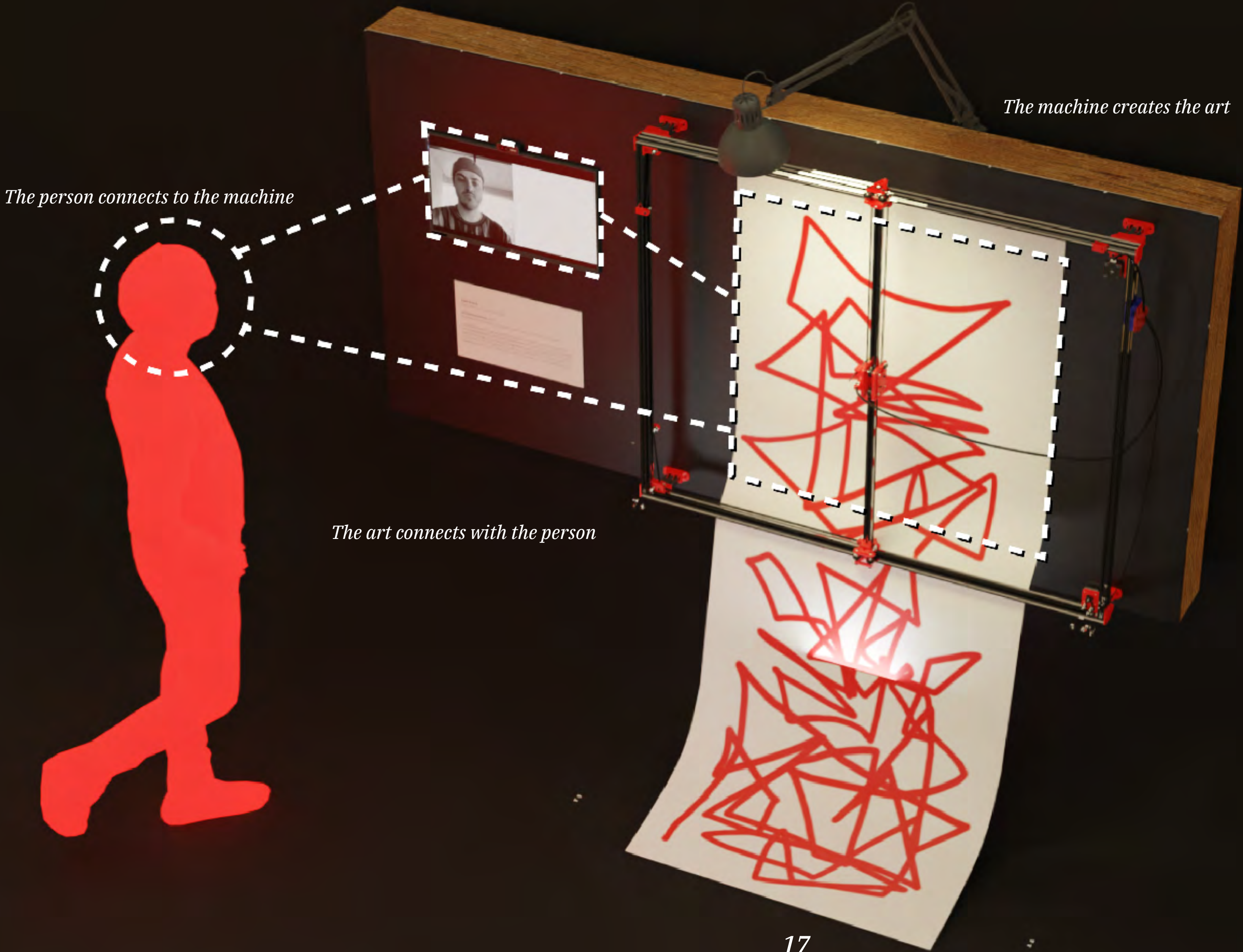
FINAL VERSION: *face_detection_drawer_v5.pde*



First test successfully making a drawing using CV data

The code works by establishing a serial connection with the plotter & sends strings of commands to mimic software communications from a plotting program. Once initialized, CV data is collected through a camera & is formatted into GCODE to be sent to the plotter. Every time the plotter responds that the previous coordinate's been executed, another coordinate can be sent.





Through the use of machine learning & CNC technologies, an art piece is created that is intrinsically connected to those who view it.

THANK YOU